V. Narayan Kushwaha

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Aerospace & Mechanical Engineering | Vehicle Dynamics | Controls | Robotics / AV | CAD | Rapid Prototyping

Seeking Fall 2026 co-op, Winter 2026-2027 internship, and May 2027 full-time roles in automotive engineering, vehicle dynamics, controls, robotics/autonomy, mechanical design, or manufacturing engineering.

EDUCATION

University at Buffalo, State University of New York — Buffalo, NY
B.S. Aerospace Engineering (Honors) and B.S. Mechanical Engineering (Honors) — *May 2027*
GPA: 3.3/4.0 | Dean's List: Fall 2023, Spring 2024

TECHNICAL SKILLS

Vehicle Dynamics & Controls: MATLAB, Simulink, Simscape, PID control, control design, path following, braking and steering models, suspension behavior, model-based systems engineering
Robotics, AV & Perception: ROS, LiDAR mapping, Intel RealSense, perceptual mapping, OpenCV, computer vision, drone ML, autonomous path tracking, sensor-based navigation
CAD/CAE: SOLIDWORKS, Siemens NX, Fusion 360, FreeCAD, SketchUp, Blender, ANSYS Fluent, Open3D
Manufacturing & Prototyping: FDM 3D printing, desktop CNC machining, laser cutting, 3D scanning, injection molding, vacuum forming, DfM, fixtures, jigs, tooling
Programming & Electronics: Python, C++, MATLAB, Arduino, Raspberry Pi, soldering, circuit assembly, data analysis, Excel automation, Sensor Fusion, CV

RELEVANT ENGINEERING PROJECTS

- Autonomous Vehicle Dynamics and Path Following: Built MATLAB/Simulink/Simscape models for the Baja SAE UB 2025 vehicle's suspension study, and braking response, steering behavior, and control-system evaluation (simulation). This helped our team make better visuals and graphs for the design report to show iterative design.
- LiDAR Mapping and Perception: Worked with Intel RealSense LiDAR and perception workflows for mapping, spatial understanding, and autonomous-navigation studies using ROS/OpenCV concepts.
- Drone ML and Path-Following System: Developed machine-learning and computer-vision workflows for UAV path-following and perception, connecting sensing outputs to navigation/control behavior at the UB DREAM Lab for a student workshop.
- Clinic Customer Tracking with OpenCV: Designed a computer-vision system to track people entering and exiting a facility for headcounting, staffing insight, and security without modifying CCTV infrastructure, using Python, any DVR Output and a bash automation on windows.
- Pipe Flow Equation and Pipe Network Optimization & Numerical FEA and Heat-Conduction Solvers: Developed MATLAB-based finite element and steady-state heat-conduction solvers for design optimization.

EXPERIENCE

Lead Staff, SEAS DREAM Lab — **March 2024 - Present**
University at Buffalo — *Buffalo, NY*
- Lead daily makerspace operations supporting students, research teams, and engineering clubs across rapid prototyping, fabrication, and troubleshooting.
- Support FDM 3D printing, desktop CNC machining, laser cutting, electronics, 3D scanning, CAD, and design-for-manufacturing workflows.
- Maintain and troubleshoot Prusa MK4S, MK3S, MK3, and MINI printers to improve print reliability, uptime, and student project turnaround.
- Designed and deployed an NFC-based print-log and email-notification system to reduce queue delays, print-failure delays, filament waste, and extruder damage.
- Conducted 30+ workshops, lab tours, and tabling events covering Arduino fundamentals, CAD workflows, 3D printing, and hands-on prototyping.
- Designed and fabricated lab tools, holders, fixtures, jigs, and rapid prototypes to improve equipment usability and workflow efficiency.

Tutor, UB Tutoring and Academic Support Services — **Jan 2024 - Present**
University at Buffalo — *Buffalo, NY*
- Tutored 200+ students one-on-one and online in General Chemistry, Physics I-II, and Calculus I-III.
- Used MATLAB, Desmos, PhET, and structured problem-solving methods to explain mechanics, calculus, and physical-system behavior.
- Created automated tutoring summaries, LaTeX-based cheat sheets, and course-material review workflows to improve student retention and study efficiency.

Engineering Intern — **May 2025 - Aug 2025**
Sketch Consultants, Architectural Consultancy Firm — *Lucknow, India*
- Developed C/C++ executables to automate beam-load calculations and structural-analysis workflows to reduce redundant hand calculations by the Structural Engineers at the firm.
- Designed cabinets, assemblies, shelves, and equipment enclosures in FreeCAD using DfM principles and client requirements.
- Modeled steady-state heat conduction through building materials to support ceiling-insulation and thermal-performance recommendations.
- Conducted site inspections and coordinated with contractors to validate CAD models, projections, BOM records, and client renders.

Researcher, Composite Materials Research Laboratory — **Oct 2023 - Nov 2024**
University at Buffalo — *Buffalo, NY*
- Investigated electrical permittivity and resistivity behavior of boron-fiber composite materials; co-authored conference work presented at MS&T 2024 in Pittsburgh, PA.

Primary Thermal Engineer, NASA-ASU L'SPACE Mission Concept Academy — **Jan 2024 - Apr 2024**
Arizona State University — *Online*
- Led spacecraft thermal modeling, heat-balance analysis, subsystem thermal trade studies, and Preliminary Design Review documentation

LEADERSHIP, CERTIFICATIONS AND AWARDS

- Structures Team, AIAA MicroG-Next: Performed heat-transfer analysis for astronaut splashdown buoy electronics and supported proposal documentation for the Preliminary Design Review.
- **Certifications:** MATLAB Machine Learning Onramp; Regression with Deep Learning; Computer Vision Onramp; Simulink Control Design Onramp; nTopology Generative Design; Siemens NX Xcelerator Academy; Digital Manufacturing & Design; Remote Sensing & GIS; CPR/AED.
- **Awards:** First-Year Engineering Design Award; MLH Best Hardware Hack